

NITYA LAGADAPATI (she/her)

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Objective: Wet-lab focused biomedical engineer capable of integrating machine learning algorithms and computational tools into wet lab workflows to guide experimental decisions, streamline hypothesis testing, and accelerate scientific discovery.

Education

Johns Hopkins University | Baltimore, MD

M.S.E Biomedical Engineering: Translational Cell and Tissue Engineering | 08/2025 - 05/2026 | GPA: 3.97/4.0

University of Washington | Seattle, WA

B.S. Bioengineering: Data Science | 09/2021-06/2025 | GPA: 3.89/4.0 (*Cum laude*)

Honors: Hershel Graves Endowed Scholarship in Engineering, Mary Gates Research Scholarship, Dean's List (2021-2025)

Research & Industry Experience

Graduate Researcher | Wilson Lab | Baltimore, MD | March 2026- Present

- Developing a polymer-antigen conjugate-based influenza vaccine platform that targets antigen-presenting cells to enhance immune response efficacy
- Utilizing downstream development pipelines to purify, characterize, and validate product quality

Graduate Researcher | Kathuria Lab | Baltimore, MD | December 2025- Present

- Processing and analyzing electrophysiology data from organoid models of the brain that assess both disease pathology and therapeutic potential

Undergraduate Researcher | Tissue and Regenerative Engineering Lab | Seattle, WA | October 2023 - July 2025

- Designed and developed a hyaluronic acid-based hydrogel system that mimics viscoelastic properties of the native meniscus for translational tissue engineering applications
- Conducted swelling, compression, and stress relaxation tests on iteratively optimized hydrogel formulations and developed MATLAB code to streamline raw data processing
- Authored experimental protocols and presented findings at 5 research conferences and symposia

Downstream Development Intern | Partner Therapeutics | Lynnwood, WA | June 2024 - September 2024

- Supported development-scale purification process optimization through high-throughput solvent screening using 96-well plate chromatography systems with the goal of transitioning the organic solvent-based process to be aqueous
- Narrowed down successful conditions by repeating studies on a larger scale and recommended solvents equivalent to current product purification standards for further studies to support safer and more sustainable process development

Formulation Development Intern | Lundbeck Seattle Biopharmaceuticals | Bothell, WA | June 2023 - September 2023

- Designed and executed controlled experimental studies evaluating the impact of pH and excipient selection on drug product stability while operating industry standard tools in a regulated laboratory environment
- Characterized and implemented UNCLE equipment for protein aggregation analysis and authored a standard operating procedure for routine use

Additional Experience

Afterschool STEM Instructor | Johns Hopkins University | Baltimore, MD | September 2025- Present

- Creating and delivering engineering design-based curricula emphasizing problem-solving and experimental thinking

Co-President | Bioengineers Without Borders | Seattle, WA | February 2022 - June 2025

- Led multidisciplinary teams across five subprojects and coordinated external collaborations
- Ideated outreach events focused on STEM education for K-12 students; executed fall/spring recruitment events yearly

Selected Projects

Blink Detection Algorithm | September 2025 – Present

- Developing and refining a blink detection algorithm to support clinical tool development at the Johns Hopkins Center for Language and Speech Processing, aimed at early diagnosis of neurodegenerative diseases

Synthesized Breath Sounds | January 2025 - June 2025

- Collaborated with ML/AI experts and medical professionals to build and optimize generative machine learning models to produce breathing sounds tailored to patient profile inputs for integration into medical training manikins

Technical Skills

Engineering/Lab: Hydrogel fabrication, Mechanical analysis, Chromatography (SEC/HPLC), CAD, SDS-PAGE, Cell Culture

Software: Python, MATLAB, R, SQL, Java, Machine Learning/Deep Learning, SolidWorks, JMP, GraphPad Prism, Origin